

CLASS 8:

FLOWCHARTS – WHILE LOOPS

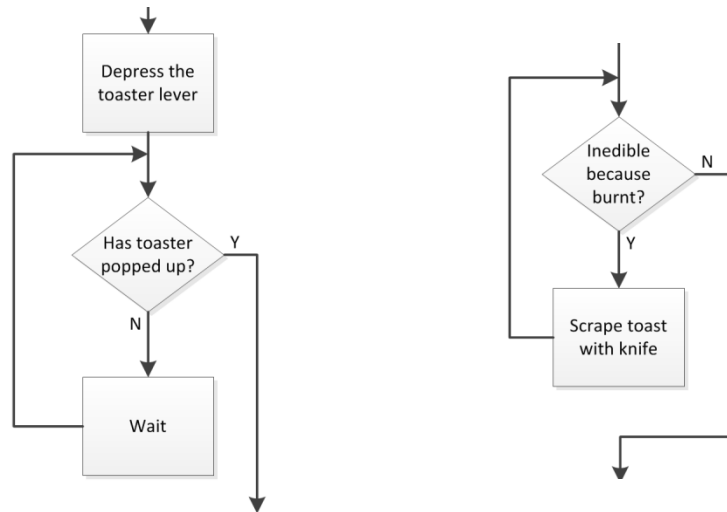
Loops

- *while* loops
- *for* loops

Loops

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- We've already seen some examples of flow charts that contain **loops**:



- Structures where the algorithmic flow loops back and repeats process steps
 - Repeats as long as a certain condition is met, e.g., toaster has not popped up, toast is inedible, etc.

Loops

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- Algorithms employ two primary types of loops:
 - ***while loops***: loops that execute as long as a specified condition is met – loop executes as many times as is necessary
 - ***for loops***: loops that execute a specified exact number of times
- Similar looking flowchart structures
 - for loop can be thought of as a special case of a while loop
 - However, the distinction between the two is very important

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while Loop

while Loop

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- Repeatedly execute an instruction or set of instructions as long as (*while*) a certain condition is met (is *true*)
- Repeat A *while* X is true
 - As soon as X is no longer true, *break* out of the loop and continue on to B
 - A may never execute
 - A may execute only once
 - A may execute forever – an ***infinite loop***
 - If A never causes X to be false
 - *Usually* not intentional

